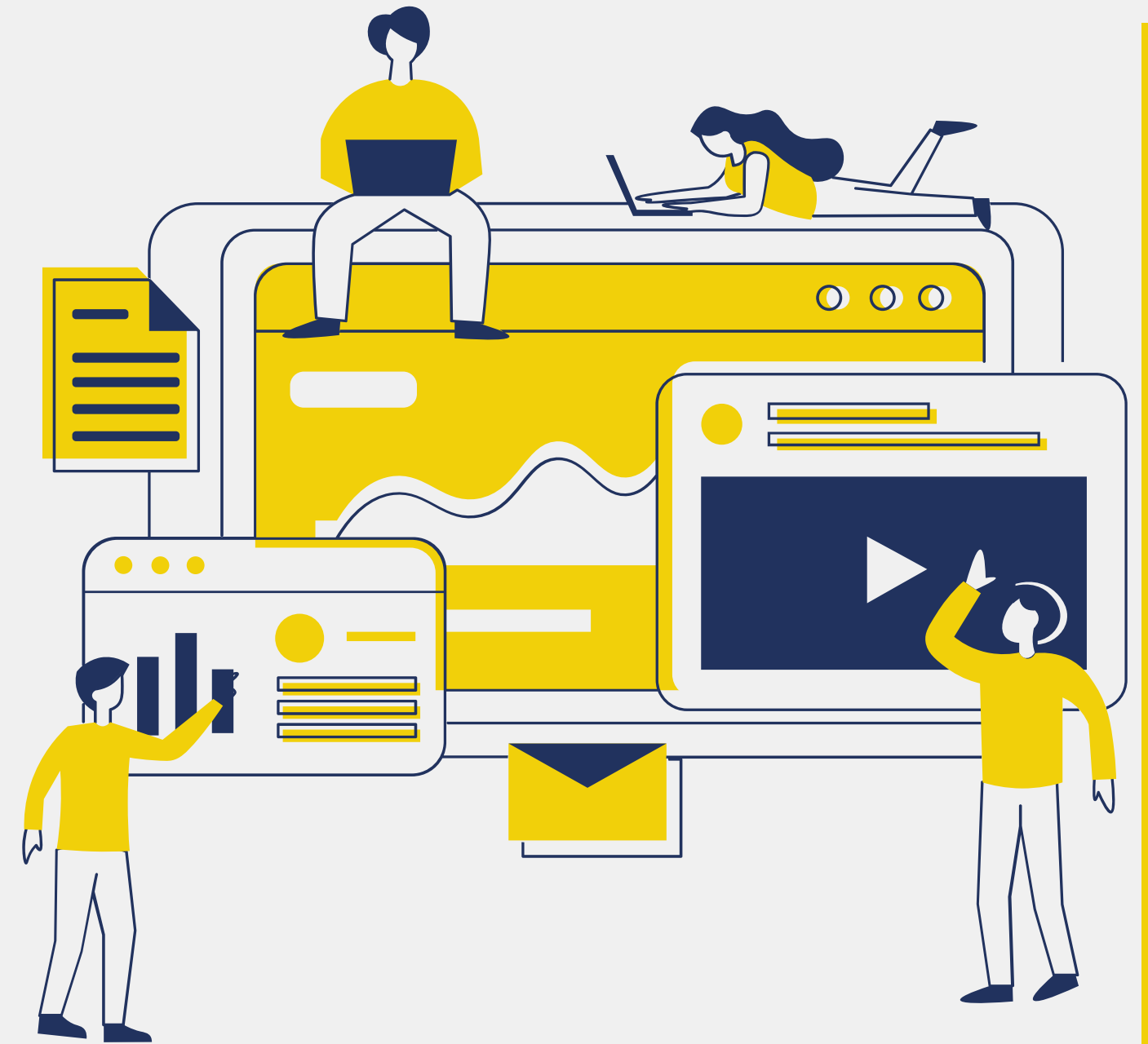




INFORMATION AND COMMUNICATION TECHNOLOGIES

Abdulkarim Abdulrakhmanuly



ABOUT ME

ABDULKARIM
ABDUL RAKHMANULY

- KAZAKH LANGUAGE AND LITERATURE TEACHER
- EDTECH SPECIALIST
- HEAD OF DISTANCE EDUCATION DIVISION (MOODLE, WEBEX, TURNITIN, PROCTORING, STUDIO)

Education:

- The University of Manchester (2020 - 2021) MA Digital Technologies, Communication and
- Al-Farabi Kazakh National University (2014 - 2016) Study of Literature, Master of Human Sciences
- Abai Kazakh National Pedagogical University (2010 - 2014) Kazakh Language and Literature

Certificates:

- Microsoft Educator Community Influencer MICROSOFT | 2018
- In-service Training Program for the Pedagogical Staff of the Republic of Kazakhstan (Level 3) the Center of Excellence of the AEO "Nazarbayev Intellectual Schools" (CoE), Almaty | 2016
- Theoretical and methodological problems of modern philology UNIVERSITY OF PÉCS, Hungary | 2015



ABOUT COURSE

01

Prerequisites: studying of discipline is based on the knowledge gained at high school on fundamentals of informatics and mathematics.

02

Language: English

03

Lectures: 15

04

Tasks: Lab works & Project work



An illustration featuring a large, glowing yellow lightbulb with a dotted texture. Two stylized figures, a woman on the left and a man on the right, are looking up at the lightbulb. The woman is wearing a blue dress and the man is wearing a blue shirt and black pants. They are standing on a dark blue, textured ground with some yellow and blue plants. A large, dark blue circle is positioned above the lightbulb, containing the text 'COURSE DESCRIPTION' in yellow. The background is white.

COURSE DESCRIPTION

ICT use in the classroom is important for giving students opportunities to learn and apply the required 21st-century skills and understand the rapid change of ICT in a technology-based world. The course allows students to master their professional and personal competencies focusing on the usage of modern information and communication technologies in various fields of professional activities, scientific and practical work, for self-education and other purposes. In addition, the the course combines theoretical and practical studies aimed at the ability to use general software applications, word processors, spreadsheets, databases, interactive presentation tools, e-mail, web browsers, website design and others. Learners will analyze, design, implement, test and evaluate ICT systems, ensuring that they are fit for purpose. The skills learnt from this course will be useful for them in their work across the curriculum, and will help to become digitally literate individuals.

REQUIREMENTS FOR THE COURSE



- COMPUTER/TABLET AVAILABILITY
- BASIC COMPUTER SKILLS
- ABILITY TO USE THE INTERNET

Thematic plan of course

WEEK 1

ICT's role in key sectors of development of society.
Standards in the field of
ICT.

Understand ICTs purposes and capabilities.
Describe ICT Topology.
Distinguish the difference between ICT and IT;
Identify attributes and limitations of different ICTs;
Identify ICT's role in key sectors of development of
society;

Practical activity: Lab work with ICT tools in education
such as Office 365, Google & Microsoft Apps, Cloud
services, etc.

WEEK 2

Google workspace in education

Understand Google workspace features and
their affordance;
Understand Google workspace's affordances in
education;
Identify Google workspace for education;
Explain collaboration options in the Google
workspace;

Practical activity: Lab work with Google Docs

WEEK 3

Microsoft products in education

Understand Office 365 features and their
affordance;
Understand Office 365 affordances in education;
Identify Office 365 for education;
Explain collaboration options in Office 365;

Practical activity: Familiarize yourself with
Windows 10 features.

WEEK 4

LMS (Moodle, Blackboard, Canvas)

Understand LMS (Learning management systems)
features;
Understand LMS elements;
Distinguish between Moodle, Blackboard and Canva
interface and difference;
Learn development stages of the courses in LMS;

Practical activity: Developing courses in LMS

Thematic plan of course

WEEK 5

Online and distance education instruments. Gamification

Understand Online and distance education principles;
Determine the main functions and parameters of instruments for online and distance education;
Describe the main aspects of Transactional distance;
Get introduced to Gamification elements in online education;

Practical activity: Developing courses in LMS and adding Gamification elements

WEEK 6

Videoconferencing tools

Understand types of videoconferencing platforms;
Define the importance of videoconferencing tools in online education;
Determine ways to provide online classes via different tools;
Get introduced to asynchronous and synchronous types of lectures via videoconferencing tools;

Practical activity: Create or provide tutorials or short lessons via videoconferencing tools

WEEK 7

Video hosting platforms and educational video types and their production

Understand the process of educational video production and its types;
Understand the types of video hosting platforms;
Learn the principles of producing educational videos on different social media platforms;

Practical activity: Produce tutorials or short lessons, educational videos

WEEK 8

Teacher collaboration and social media

Understand methods and structures of cloud systems;
Understand the instruments for online teacher collaboration;
Define the advantages of online collaboration for teachers;
Determine basic organizational methods in teacher collaboration;

Practical activity: Find colleagues via social media and collaborate with them in creating an educational video

Thematic plan of course

WEEK 9

Website construction platforms

Understand website construction platforms;
Demonstrate advantages of the website for teacher community collaboration;
Understand the affordances of websites for sharing educational products online;

Practical Activity: Learn to design websites with the WIX constructor

WEEK 10

Online platform for teachers' development

Define online professional development options for teachers;
Understand high-tech companies' educational certification systems and initiatives for the teacher community;
Determine a basic understanding of opportunities for teachers for professional development;

Practical activities:
Learn to work with Office 365 Compare Google Drive/ One Drive/ Yandex Disk/ Dropbox opportunities

WEEK 11

Multimedia technologies

Identify the essential features of graphics/image data types and file formats;
Define multimedia data representations tools;
Compare multimedia technologies for planning, business processes and their visualization;

Practical activities: Learn to create interactive presentations with Prezi; Learn graphic design platform Canva; Learn to visualize data with Powtoon

WEEK 12

Cybersecurity

Identify computer system threats and evaluate their impact;
Understand the nature of secure software development, operating systems and database design;
Understand the security management methods to maintain security protection;
Define the effectiveness of various cryptographic techniques;
Determine basic organizational security policies;

Practical activity: Encryption/decryption methods

Thematic plan of course

WEEK 13

E-technologies. Electronic business. Elearning. Electronic government.

Understand concepts of E-business: main models of electronic business;
Learn how to create and manage electronic textbooks with Kotobee Author & iBook Author;
Understand electronic government: concept, architecture, services Learn to work with egov.kz
Define how e-government is developing in Kazakhstan

Practical activity: Lab work with egov.kz

WEEK 14

Information technologies in the professional sphere. Industrial ICT.

Identify software for the solution of tasks of the specialized professional sphere;
Determine modern IT trends in the professional sphere: education, medicine, etc.;
Compare search engines and electronic resources for professional purposes;
Describe safety issues in industrial ICT;

Practical activity: Learn to apply Office 365 office products in an education

WEEK 15

Perspectives of development of ICT ;

Identify prospects of development in the sphere of the IT market;
Define forming of an ecosystem of IT;
Identify perspectives of startups;
Determine the development of necessary infrastructure of electronic payments and logistics;
Identify prospects of development of Etechnologies;

Practical activity: Project Work

WEEK 16

Fianls



Lab work Outcomes

**AT THE END OF THE COURSE YOU'LL BE ABLE TO EASILY
MANAGE THE FOLLOWING TOOLS AND BE DIGITALLY LITERATE**

1. Google Apps: Google Classroom (alternative: Edmodo)
Google Drive (alternative: YandexDisk, Dropbox, OneDrive, etc.)
Google Doc (alternatives: Microsoft Word, Pages)
Google Sheet (alternative: Microsoft Excel, Numbers)
Google Slide (alternatives: Power Point, Keynote)
Google Form (alternative: Microsoft Form, etc.)
2. Design tools: Canva, Prezi, Figma, Visme, etc.
3. Gamification tools: Kahoot, Quizlet, Quizizz, etc.
4. Project Management tools: Trello, Asana, Jira, Microsoft Project, etc.
5. Video creators/editors: EdPuzzle, Powtoon, Vyond, Animaker, etc.
6. Email marketing tools: Mailer lite, Mailchimp, Mailjet, etc.
7. Web site constructors: Tilda Wix, etc.
8. E-learning platforms: Udemy, Coursera, EdX, SkillShare, Skill.im, etc.
9. E-book creators: Scrivener, iBooks Author, iSpring, etc.
9. Network: Cisco
10. Document preparation system: LaTeX
11. Governmental portal: egov.kz



ONLINE CLASS RULES

LECTURE MATERIALS WILL BE REGULARLY UPLOADED TO THE MOODLE, WHICH MEANS WE WILL NOT BE MEETING ON WEBEX. YOU'RE FREE TO DECIDE WHEN TO READ AND GET INTRODUCED TO LECTURE MATERIALS.

MOODLE.

HW/LAB WORK GRADES WILL BE AVAILABLE IN MOODLE.

FOR EACH PRACTICE WORK WILL BE WRITTEN COMMENTS, DO NOT FORGET TO READ THEM.

ALL HW/PRACTICE WORKS ARE TO BE SUBMITTED ON THE DAY THEY ARE DUE. STUDENTS WILL BE PENALIZED 10% (OUT OF 100%) FOR EVERY DAY A PW IS LATE. HW/PW WORKS SUBMITTED OVER ONE WEEK AFTER THE DEADLINE ARE NOT ACCEPTABLE! IF YOU BECOME ILL OR THE VICTIM OF AN EMERGENCY, CONTACT ME WITHIN 48 HOURS.

COPIED AND PASTED HW/PW WILL NOT BE CONSIDERED/ACCEPTED AT ALL! IN THIS CASE BOTH STUDENTS (THOSE WHO GAVE THE WORK TO COPY AND THE ONE WHO HAVE COPIED) WILL GET 0.

NO MAKE-UP TESTS/QUIZZES/MIDTERM EXAMS ARE ALLOWED UNLESS THERE IS A VALID REASON FOR MISSING THEM.

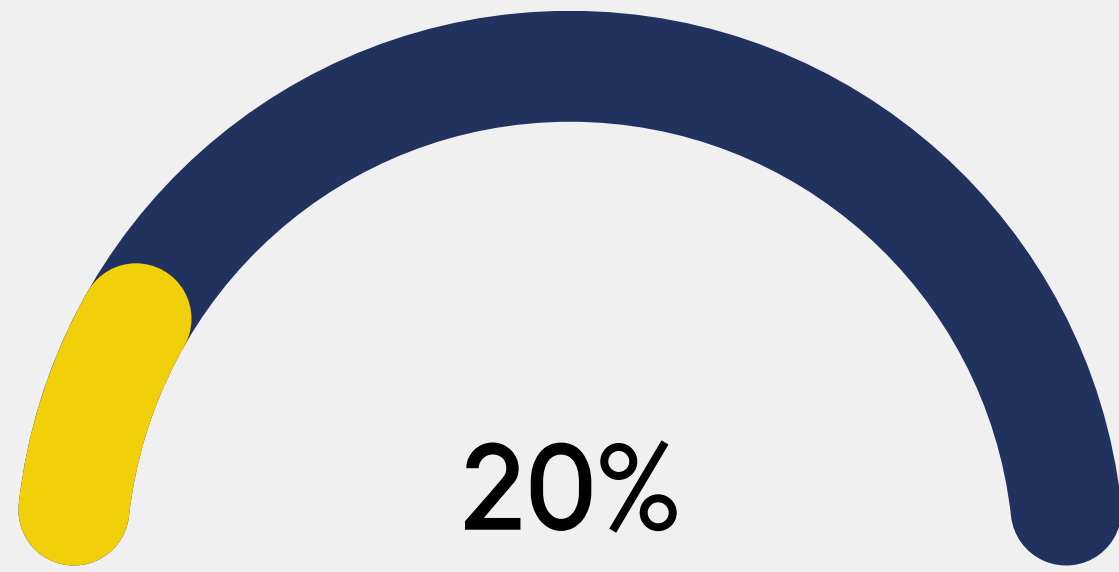
ALL HW/PW SHOULD BE SUBMITTED TO THE MOODLE, OTHER PLATFORMS WON'T BE CHECKED. IF YOU MISSED DEADLINE (BASED ON CONDITION 1), ONLY, IN THIS CASE, HW/PW WILL BE ACCEPTED THROUGH GMAIL.

WHILE TEXTING YOUR INSTRUCTOR USE ONLY CORPORATIVE GMAIL ACCOUNT (STU.SDU.EDU.KZ).

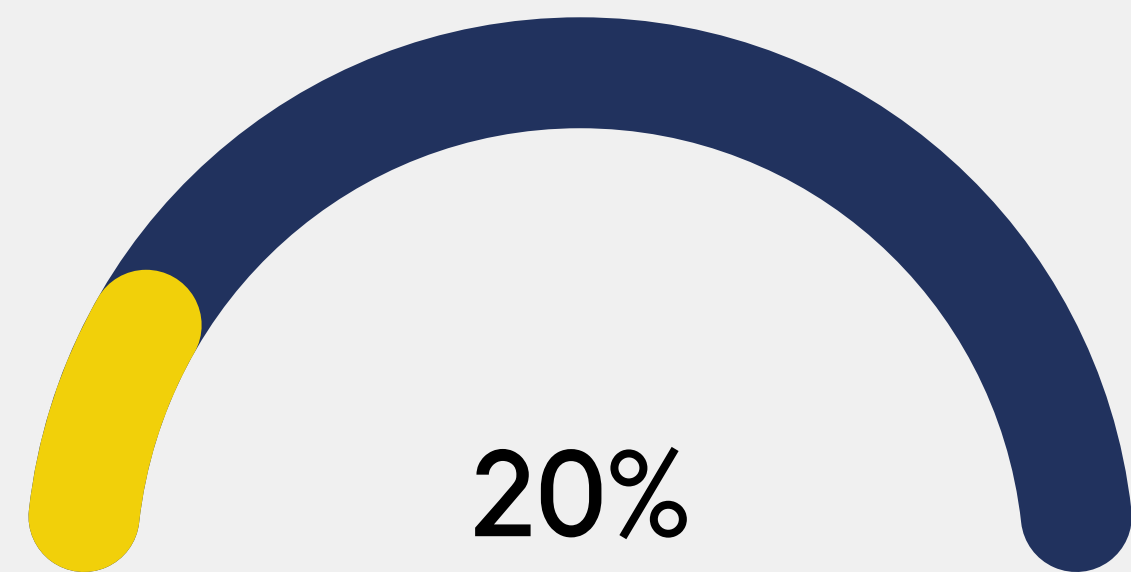
WHILE SENDING GMAIL MESSAGES MAKE SURE THAT EVERYTHING IS CLEAR.

TEXT ME BETWEEN 9AM-6PM!!! ALL GMAIL MESSAGES WILL BE ANSWERED MAX WITHIN 2 DAYS.

ASSESSMENT METHODS AND CRITERIA



Fast typing (40-50-60wpm)

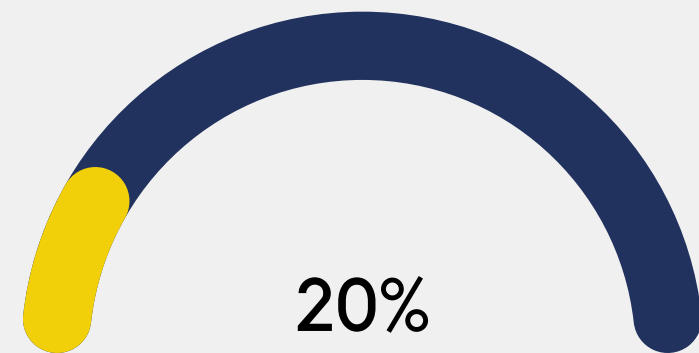


Get certification from Google

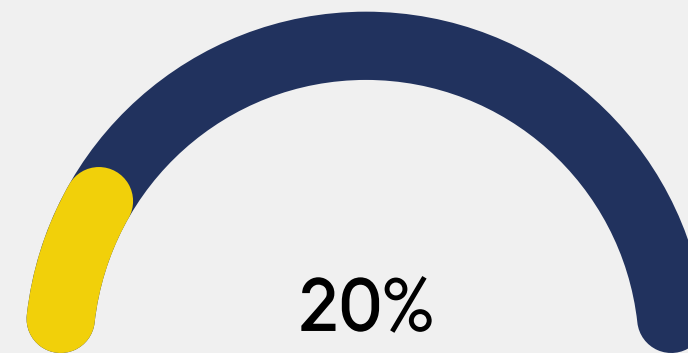
The University's normative rules regarding assessment apply. See the Code of Practice on Assessments.

These norms set the boundary conditions for all instructors of University.

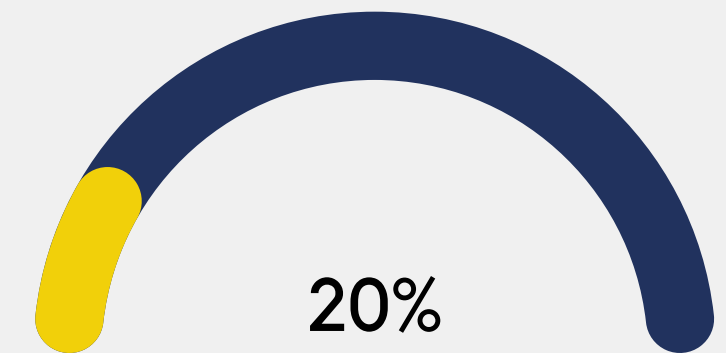
ASSESSMENT METHODS AND CRITERIA



Practice works
based
on lecture
topics (10 PW)



Quiz (2)



Participation

The University's normative rules regarding assessment apply. See the Code of Practice on Assessments.

These norms set the boundary conditions for all instructors of the University.





.....> **THANK YOU** <.....